# EcoBin

Autonomous **Source-Segregation** & Community-Scale Bio-Utility

***"Throw Waste. Create Energy."***



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INTEGRATED EDGE-AI WASTE SORTER WITH SOCIETY & CANTEEN ENERGY RECOVERY

# Structural Breakdown in Urban Waste Logistics

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## Municipal Haulage Overhead

**Cities expend 50% to 70% of solid waste budgets purely on diesel and fleet logistics to haul heavy, wet food scraps to distant landfills.**

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## Material Degradation

**Mixing wet food scraps with dry packaging permanently spoils over 80% of recyclable plastic, cardboard, and paper value.**

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## Biogas Digester Failure

**Communal and institutional anaerobic digesters choke and fail because humans inadvertently mix plastic films into food waste.**

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## Systemic Insight

**The crisis is not a lack of processing plants, but the** failure of automated segregation at the source.

# Automated Segregation at Source, Decentralized Energy at Scale

## Stage 1 — Tower & Canteen Ingestion

**Standalone Smart Segregation Nodes on residential floors and dining halls that automatically isolate clean dry recyclables from wet organics.**

## Stage 2 — Central Utility Integration

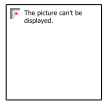
**Pure organic slurry flows into a centralized 100–300 kg/day society digester located in the utility yard or basement.**

## Direct Outputs

**Clean baled dry recyclables, localized cooking biogas for community messes, and pathogen-safe organic fertilizer.**

**Product Evolution: EcoBin Family (residential homes) → EcoBin Society (high-rise complexes & institutional canteens)**

# Embedded Edge-AI & Closed-Loop Bio-Engineering



## Perception Unit

**ESP32-S3 microcontroller executing INT8-quantized YOLOv8 object detection in under** 200 milliseconds.



## Multi-Sensor Fusion Array

**Combines optical bounding with capacitive moisture, inductive metal, and weight sensors to achieve** >90% sorting accuracy **on soiled trash.**



## Mechanical Pre-Treatment

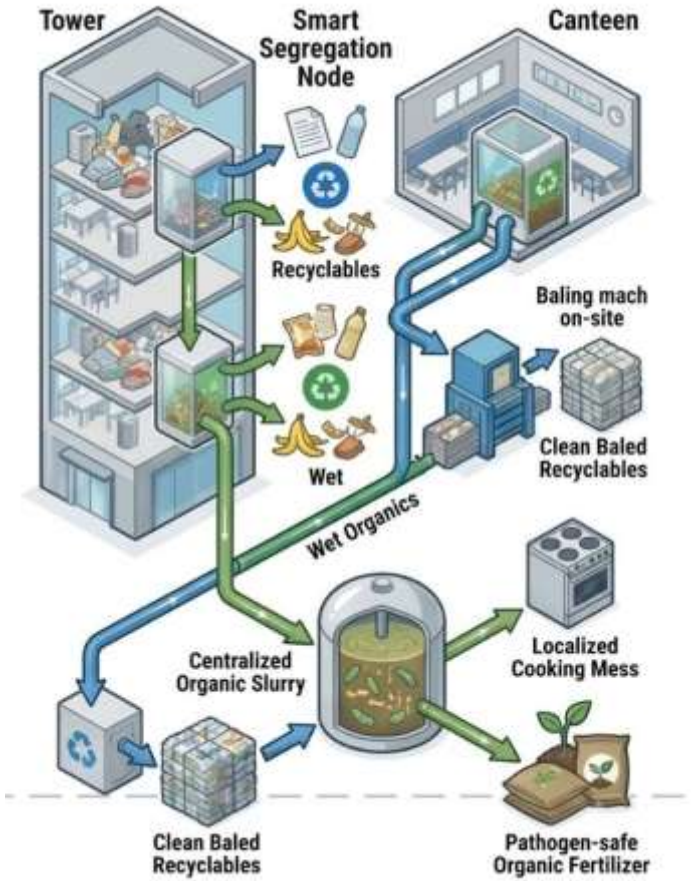
**High-torque stainless-steel macerator shears organic scraps into a** <2 mm micro-slurry **with automated water micro-dosing to accelerate hydrolysis.**



## Active Anti-Souring & Safety

**Real-time pH telemetry auto-doses** $NaHCO_3$ **buffer. Continuous activated carbon scrubbers eliminate odors; fail-closed solenoids seal gas lines if methane exceeds** 0.5%.

# B2B Beachhead: Gated Societies & Institutional Canteens




## High-Density Target Segments

- **1,200+ Gated Residential Societies (HOAs)** facing municipal penalties for not processing wet waste on-site.
- **University Hostels, Corporate Canteens, and Commercial Food Courts** generating 100–300 kg of daily bulk kitchen scraps.

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## Defensible Value Proposition

- Slashes third-party private waste-hauling contracts by **50% to 60%.**
- Replaces commercial LPG expenses for society clubhouses, canteens, or maintenance staff messes.
- Provides high-purity dry recyclable bales sold at premium rates to industrial recyclers.

# Commercial Revenue Streams & Rapid ROI

## Hardware Sales (CapEx)

EcoBin Family: ₹40,000 – ₹60,000 for villas and individual residences.

EcoBin Society: Modular community hubs ₹3 Lakhs – ₹8 Lakhs scaled to flat count and canteen volume.

## Recurring AMC (OpEx)

₹5,000–₹10,000/month for bacterial culture refills, buffer replenishments, and filter servicing.

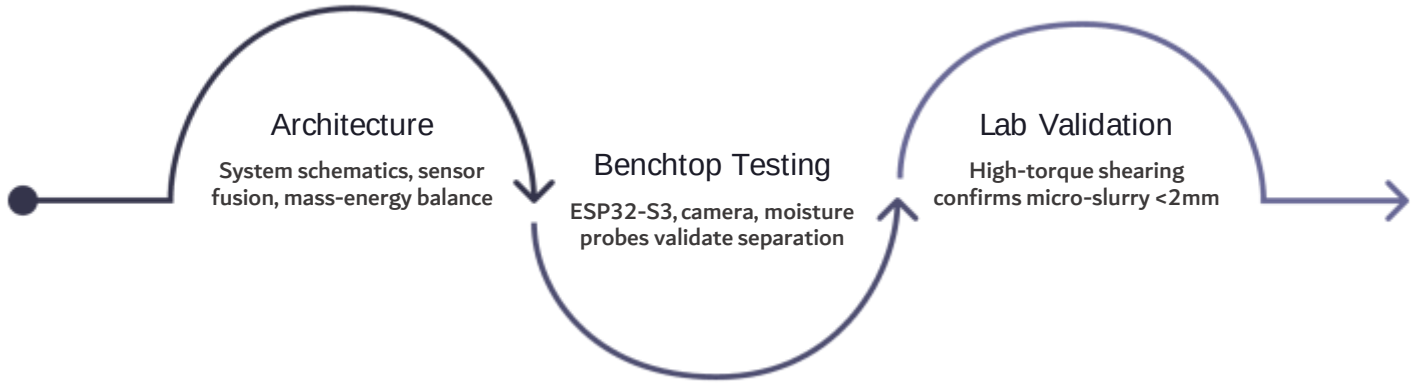
## Recyclable Offtake

Direct revenue from selling uncontaminated dry recyclable bales (PET, aluminum, cardboard) to recycling partners.

## Customer Payback

Gated societies recover initial investment within 18 to 24 months through hauling fee reductions and energy savings.

# System Architecture Formulation & Validation Design




## Architectural Readiness

- System schematics, sensor fusion logic trees, and mass-energy balance calculations fully mapped out.
- Mechanical layout designed for safe communal deployment in utility yards, avoiding indoor residential gas storage risks.

## Validation Objectives

- Benchtop testing rig integrating ESP32-S3, camera, and moisture probes to prove real-time plastic vs. organic food separation.
- Laboratory testing of high-torque shearing to confirm uniform micro-slurry reduction (<2 mm).

Leadership: **Akshi Rani (Founder & System Lead) managing AI pipelines and product architecture, partnering with campus engineering labs for physical assembly.**

# Phased Engineering & Pilot Milestones

1

## Phase 1 — Months 0–3

**Benchtop validation of camera and multi-sensor segregation logic on mixed waste in college lab.**

2

## Phase 2 — Months 3–6

**Lab-scale 20L testing for mechanical maceration kinetics and automated pH buffer calibration.**

3

## Phase 3 — Months 6–12

**Prototype pilot deployment at AKGEC hostel canteen with real-world bulk food waste.**

4

## Phase 4 — Months 12+

**Industrial design for manufacturing (DFM), modular scaling, and commercial deployment across gated housing societies.**

# Initial Prototype & Bench Validation Budget

TOTAL GRANT ASK: ₹2.5 LAKHS

**Pre-Incubation Prototype & Validation Grant**



40%

## Development Hardware

ESP32-S3 boards, camera modules, industrial moisture/metal sensors, servo actuators — **₹1,00,000**

30%

## Mechanical Fabrication

Acrylic drop chute, small test macerator blade assembly, and laboratory test enclosures — **₹75,000**

20%

## Lab Testing & Safety

Reagents, pH calibration solutions, and safety isolation valves — **₹50,000**

10%

## Miscellaneous

Wiring, testing consumables, and analytical reporting — **₹25,000**

# Decentralizing Urban Waste Management

> "Transforming community waste from an escalating logistical expense into clean decentralized energy and high-purity resources right at the source."

## Presented By

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## Seeking

₹2.5 Lakhs Pre-Incubation Funding

Prototype & Bench Validation Grant

## Next Step

Thank You! Open for Technical Evaluation and Discussion.